

PHIX

Hospital Maintenance App *Comprehensive Feature List*

55 features across 11 categories

Security & Access Control

- User registration via short-lived, admin/manager-generated 6-digit PIN codes, scoped to a specific role
- Email/password authentication with mandatory email verification before login is allowed
- Resend verification email flow directly from the login screen
- Forgot password / self-service password reset via email link
- Login attempt limiting with a 5-attempt lockout and 30-second cooldown (persisted across refreshes)
- Account deactivation/reactivation and permanent deletion by Admin/Manager
- Admin-triggered password reset emails for any user

Backend & Data Security

- Google reCAPTCHA-backed Firebase App Check enforced on sensitive backend API routes, blocking requests that don't originate from the real app
- Registration ID (PIN) claiming handled server-side via the Firebase Admin SDK, with per-user and per-IP rate limiting to prevent brute-force PIN guessing
- Granular Firestore security rules enforcing exactly which fields each role can read or write, per collection and per report status
- Protected route wrapper with real-time session enforcement — a deactivated or deleted account is signed out immediately via a live profile listener, not just on next login

Onboarding & Account Management

- Multi-step (3-step) sign-up form with progress dots for account, contact, and role/registration details
- Role-based accounts and routing: Staff, Worker, Estate Manager, Admin, Manager, Procurement
- Push notification device linking on login (Median/OneSignal)

Report Submission & Intake

- Staff maintenance report submission form (category, priority, location, description, optional photo)
- Client-side photo compression before upload, with file-size and file-type validation
- Configurable report priority levels (Emergency, Urgent, Routine) with automatic due-date calculation
- Cancellation/deletion of a staff member's own still-incoming reports

Report Workflow & Lifecycle

- Full multi-stage report workflow: Incoming → Approved/Denied → Materials Request → Confirmed → Procured → Assigned → Accepted/Rejected/Dropped → Completed → Closed/Reopened
- Admin approval/denial of incoming reports, with denial reason notes

- Estate Manager materials/cost request submission and admin confirmation or denial of costs
- Procurement stage for marking materials as purchased with recorded cost
- Technician (worker) assignment and reassignment by Estate Manager, including handling of rejected and dropped jobs
- Worker accept/reject flow for assigned jobs, with a mandatory reason captured on rejection
- Worker “drop job” action for accepted/reopened work the assigned technician can no longer complete — requires a reason and automatically returns the job to the Estate Manager for reassignment
- Before/after photo capture per report, stored separately from report data for performance
- Report completion marking by assigned worker

Feedback & Quality Assurance

- Staff feedback and 1–5 star technician rating on completed work
- Staff ability to reopen unsatisfactory completed jobs, with reason capture
- Staff ability to close reopened jobs directly without a full new feedback cycle
- Soft-delete (‘dismiss’) of denied reports by the reporter, preserving records for analytics

Notifications & Real-Time Updates

- Real-time in-app notifications (‘New’, ‘Feedback’) with per-user read tracking and bulk ‘mark all as read’
- Push notifications (OneSignal) for key status transitions, targeted to the right role/user with deep links
- Automatic overdue detection and flagging for active reports past their due date
- Live countdown timers on report cards showing time remaining until due
- Emergency-overdue alert banner for at-a-glance urgent attention items

Reporting Tools & Documents

- Detailed report view panel showing full timeline, technician contact, photos, denial/reopen/drop reasons, and feedback
- Estate Manager and technician phone-call quick actions from report details
- Downloadable, professionally formatted PDF materials/work-order report per maintenance request

Analytics & Dashboard

- Comprehensive analytics Dashboard: KPIs, status/priority breakdowns, category demand, cost analytics
- Dashboard financial analytics: total/average/highest job cost, spend by category and priority, monthly spend trend
- Dashboard quality/risk metrics: denial rate, reassignment rate, first-time-fix rate, pipeline stage durations, location hotspots
- Worker performance leaderboard with blended score (customer rating, reliability/rework rate, completion speed)
- Average job resolution/completion time tracking, fleet-wide and per worker
- Recent activity feed and recent denial-reason feed for admins/managers
- Downloadable dashboard summary PDF report for a selectable period (month/year/last year/overall)

Reports Browsing & User Management

- Paginated, filterable, and searchable report lists per role with time-range filters (week/month/year/etc.)
- ‘Pending Reports’ tracking page for staff with cancel, close, and dismiss actions and status filter tabs
- ‘History’ page for staff showing all closed reports with search, sort, and resolved-vs-reopened filtering
- User management console: search, filter by role, view join dates, deactivate/reactivate/delete accounts

UI/UX & Branding

- Role-themed dashboards/home pages (distinct color themes per role: Admin, Estate, Worker, Manager, Procurement)
- System-wide light/dark mode toggle, with automatic detection of the device's OS-level preference on first visit and the user's choice persisted across sessions
- Mobile-responsive layouts with slide-in panels, bottom sheets, and mobile-specific navigation
- Hospital-branded UI with animated glassmorphism login screen and tiled watermark background
- Birthday recognition: automatic in-app birthday banner and confetti animation for staff on their birthday
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